BUCKLEYS

MANUFACTURERS OF SPECIALIST TEST EQUIPMENT

Underwater Corrosion Probes UCP1A, UCP1B & UCP1D

Operating Instruction

English 06/26



Please read carefully before use

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Description

The UCP 1A, 1B and 1D contact probes are versatile instruments which are used to carry out under water cathodic potential measurements.

The units are designed for use with manned or remotely operated vehicles (ROVs). There are three types available for use:

UCP1A - Contact Probe

Operation of the UCP1A is carried out by way of a probe tip making contact with the surface under test and reading the resulting voltage on a suitable high impedance (> 10M Ohms) digital voltmeter.

The UCP1A has two connections:

- Black Wire - Stainless Steel Probe Tip
- White Wire - Ag/AgCl Half Cell

UCP1B - Proximity Probe

Operation of the UCP1B is carried out by way of placing the unit in close proximity to the surface under test and reading the resulting voltage on a suitable high impedance (>10M ohms) digital voltmeter. The meter should be connected between the probe and a separate structure negative return.

The UCP1B has two connections:

- Black Wire - Not connected
- White Wire - Ag/AgCl Half Cell

Both types of UCP probe utilise a Silver/Silver Chloride (Ag/AgCl)/Seawater half-cell. The accuracy of the cell should be checked against a K-SAT Reference Cell (6004-0111) to ensure that it is still accurate and has not been contaminated in use. Check the condition of the UCP1A Probe Tip to ensure that it is still pointed, replace if necessary (6004-0015/6004-0017).

UCP1D – Digital Contact Probe

The UCP1D operates in a similar way to the UCP1A but is designed specifically for seamless integration into ROV systems and data logging platforms.

The UCP1D has four connections:

- Red Wire - Positive
- Black Wire - Negative

- Orange Wire – RS-485_A
- Brown Wire – RS-485_B

Unpacking

Remove the UCP from its packaging and check for damage. If any part is damaged, the carrier and supplier should be notified immediately. All packing material should be kept for inspection and the UCP should not be used.

The package will contain one of the following items depending on which UCP probe that you purchased:

UCP1A – Contact Probe, **UCP1B** – Proximity Probe, or **UCP1D** – Contact Probe

If the probe type is not what you expect please contact your supplier.

Maintenance

Replacement of electrode

Unscrew the two recessed nylon screws on either side of the body.

Carefully pull the front and rear assemblies apart.

> **Note:** Take care not to break the connecting flexible wire that connects to the Probe tip in the UCP1A/D contact probe.

Unscrew the old Half Cell.

Check that there is no moisture in the Cell's recess and clean if necessary. Check the O-ring (size 011) for damage and replace if necessary. Smear with a small amount of silicone grease and fit on the new reference cell.

Screw the new Half Cell into the body until the O-ring just bottoms, then tighten by a further half a turn.

Reassemble the unit.

> **Note:** Take care not to trap or break the interconnecting wire that connects to the Probe tip in the UCP1A/D contact probe.

Refit and tighten the two nylon screws on either side of the body.

Long Term Storage

Flush well in clean seawater to remove any contamination from the Nose Cone and allow to dry.

If salt encrustation is seen on the nose cone, remove the Ag/AgCl half-cell and briefly wash in fresh water but immediately after, thoroughly rinse in salt water (3% salinity).

Recommissioning

Soak the UCP for 2 hours in a seawater solution (3% salinity). Carry out verification checks (see page 7).

Calibration

There are three types of UCP probe, the 1A 'Contact Probe', the 1B 'Proximity Probe', and the 1D 'Digital Contact Probe'.

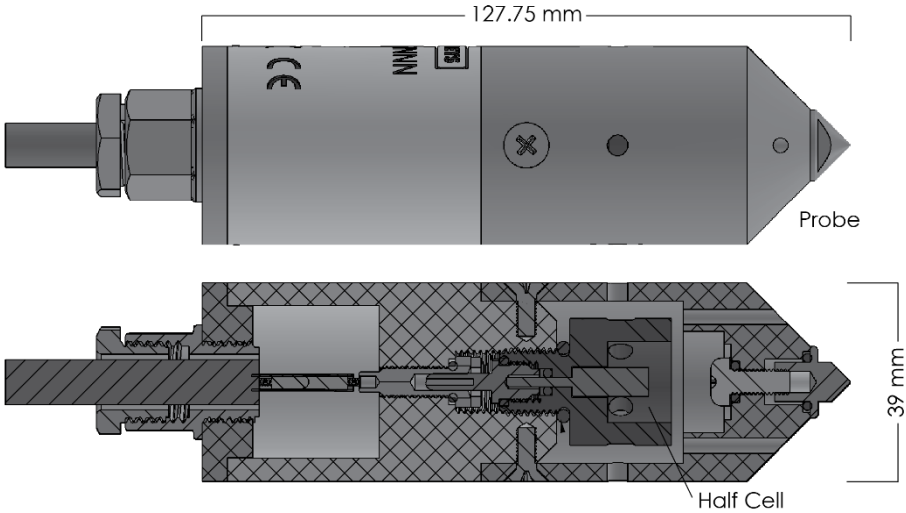


Fig.1 - UCP 1A Contact Probe

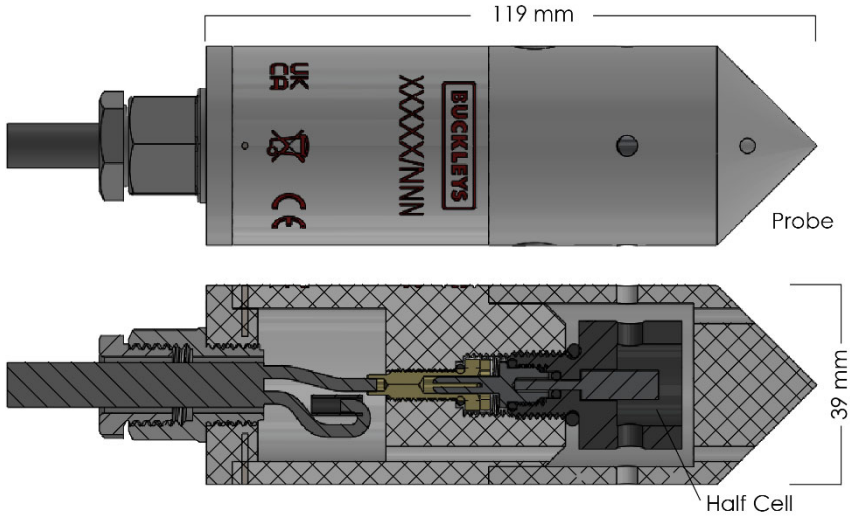


Fig.2 - UCP 1B Proximity Probe

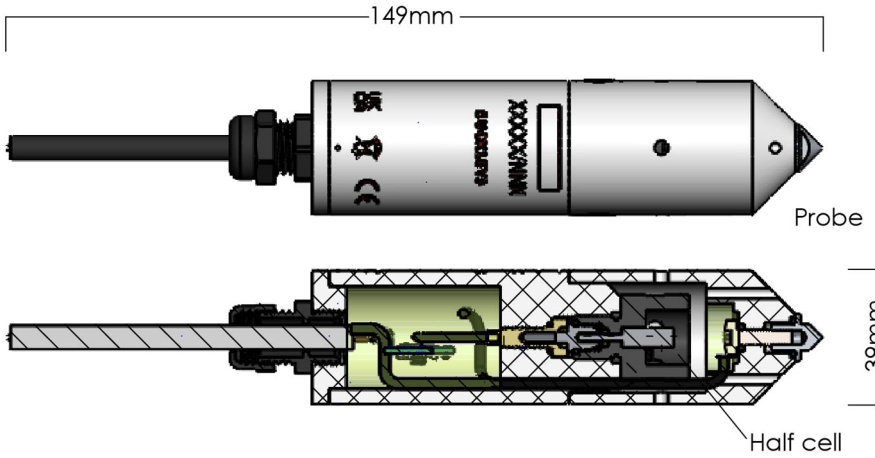


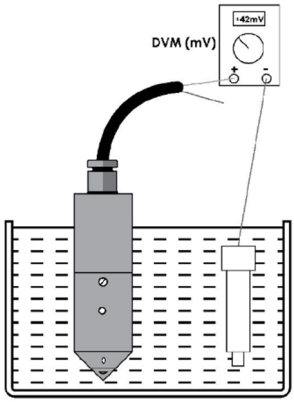
Fig.3 - UCP 1D Digital Contact Probe

All types use the same type of replaceable silver/silver chloride/seawater half-cell.

Verification of the Ag/AgCl Half-Cell (UCP1A & UCP1B only)

Plastic container filled with 3% by weight salinity artificial seawater solution.

K-SAT Reference Electrode. (Use a glass type or plastic screw in with an adaptor)



Note 1: Remember to remove the plastic cap on the plastic reference electrode).

Note 2: No bare wires should be exposed to the seawater.

Fig.4 - Half-cell verification

Screw the adaptor lead (6004-0018) onto the reference electrode (plastic screw on type). Submerge the UCP and reference electrode in the 3% seawater solution. 3% seawater should be at an ambient temperature of 25°C.

Connect the white lead from the UCP to the Positive terminal of the DVM (see Fig. 4 above).

Connect the lead from the K-SAT reference electrode to the negative terminal of the DVM.

Allow the electrodes to reach a stable temperature and potential (15 - 30 minutes).

The voltage difference between the reference electrode and the internal silver/silver chloride electrode can be read directly off the DVM display.

Measurement result

Verify a nominal potential of +54mV ± 10mV for the K-SAT cell.

Note: The values recorded for the Silver/Silver Chloride Electrode are dependent upon the Salinity and Temperature of the Seawater at the time of measurement.

Test using Zinc test blocks

Verification of the operation of the UCP1A/D can be carried out using a Zinc test block.

For UCP1A

Connect the white connection from the UCP1A to the positive terminal of the DVM, and the black connection to the negative.

For UCP1D

Connect the 4-core cable to a RS-485 port or suitable RS-485 converter and using a suitable serial terminal or datalogging software proceed as detailed below.

Soak the UCP1A/D for 2 hours in a seawater solution. The salt concentration has a great effect on the readings, so does the state of the test block.

A tarnished Zinc block will give a lower reading than clean one. Do not leave the test block in the seawater – remove it after use.

Touch the tip of the UCP1A/D to the Zinc Block while it is immersed on the 3% salt solution.

Potential measurements taken should be logged to check if any significant variation occurs. Differences in the order of 10mV between readings are quite possible and will mainly be caused by variations in water salinity at different locations or due to changes in water temperature.

As a guide, readings taken in a 3% salt solution at ambient temperature of 25°C are as follows:

Zinc (Zn) = 1.002V - 1.022V

For UCP1B

For the UCP1B connect the white lead from the UCP to the positive terminal of the mV meter and negative terminal of the meter to a zinc block.

Note: The connection to the zinc block needs to be above the waterline.

When the UCP1B is submerged in the water in close proximity to the zinc block the readings will be as shown for the UCP1A above.

The UCP1B should also be soaked for 2 hours in the 3% seawater solution prior to taking any measurements.

UCP1D Commands

The following commands to use the UCP1D to be issued as ASCII strings:

SW Returns the software revision.

BD {BAUD RATE} Selectable baud rates are: 1200, 1800, 2400, 3600, 4800, 7200, 9600, 14400, 19200, 28800, 38400, 57600, 115200

M0 Sets the unit into poll mode where a "ME" command will return the last data sample in volts.

M1 Sets the unit into poll mode where a "ME" command will return the last data sample in millivolts.

M2 Sets the unit into continuous output mode where the interval is defined by the interval "IN" command, the data returned will be in volts.

M3 Sets the unit into continuous output mode where the interval is defined by the interval "IN" command, the data returned will be in millivolts.

ME Returns the latest measurement result in volts (mode 1) or millivolts (mode 2) depending on the selected mode, this command will have no effect in mode 2/3

IN {INTERVAL MILLISECONDS} Sets the data output interval for continuous output modes, this can be set from 100 milliseconds to 60,000 milliseconds, values outside of this range will be ignored.

TE Returns the temperature of the unit's MCU in degrees Celsius, the temperature reading of the MCU will read the water temperature but only once the unit has had sufficient time to equalize in temperature with its surroundings.

Product Variations

The UCP1A is available with varying lengths of underwater cables as follows:

Cable Length	**Part Number**
3m:	6004-0039
50m:	6004-0098
100m:	6004-0099
200m:	6004-0100
300m:	6004-0101
*Free issue whip:	6004-0042

***UCP1A assembled with cable and/or plug provided by customer**

The UCP1B is available with varying lengths of underwater cables as follows:

Cable Length	Part Number
3m:	6004-0043
50m:	6004-0102
100m:	6004-0103
200m:	6004-0104
300m:	6004-0105
*Free issue whip:	6004-0050

***UCP1B assembled with cable and/or plug provided by customer**

The UCP1D is available with a 5m underwater cable:

Cable Length	Part Number
5m:	6004-0142

Disposal information

Producer registration number: WEE/HJ0051TQ



This Product must be disposed of in accordance with UK WEEE Producer Responsibility Regulations, or in accordance with your local WEEE guidance.

For further information on UK WEEE regulations click on:
http://www.gov.uk/government/collections/producer-responsibility-regulations

Contact details

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Tel: +44 (0)1303 278888
Website: www.buckleysinternational.com

Distributor details

Product registration

Thank you for choosing a Buckleys product, we are sure it will provide you with many years of reliable service.



Please register this product via Buckleys' website and download the Warranty Registration Certificate.

Register your product in 5 minutes

Once your product is registered, you will receive the following benefits:

- **FREE** annual service & calibration reminders by email
- **Latest** industry news relating to your product
- Be the **first** to hear about our new products

We strive to improve the quality of our products and service.

Registering your product helps us monitor overall quality of our products, service and dealer network. Additionally, if we ever need to contact you regarding your product, we are able to do so immediately.

We will also send you annual service/calibration reminders by email to help ensure your product is always in perfect working order.

To register your product, simply visit:

www.buckleysinternational.com/registration

...Complete the online form and click on SUBMIT.